

Shawn Guo

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Software Engineer | Systems, Gameplay, and Audio Programmer

EDUCATION

Johns Hopkins University

Baltimore MD

B.S. in Computer Science; B.M. in Music for New Media

May 2026

- Focus: Systems Programming, Audio Programming, Game Engine Programming, Computer Graphics
- Head Teaching Assistant for Game Engines Programming, Computer Graphics and 3D Game Programming
- Co-Founder and President of Hopkins Game Development Society

EXPERIENCE

Lead Engineer, Audio

BALTIMORE MD

Snow Day Games LLC

August 2025 - Present

- **Lead** audio software development for Vesuvius, a \$15K grant-funded action-adventure game, building Wwise-integrated gameplay audio architecture for state-driven ambience, combat feedback, room-aware transitions, and runtime event pipelines.

Software Engineering Intern, Audio

MIDDLETON WI

Activision/Raven Software

May 2025 - August 2025

- **Developed and maintained** low-level C/C++ runtime systems and APIs within a proprietary AAA engine, supporting scalable and data-driven gameplay audio pipelines across Warzone and Black Ops 7
- **Profiled, debugged, and optimized** real-time audio systems, focusing on runtime performance and memory behavior
- **Created** a C#/ .NET WPF-based audio asset compression tool used in production pipelines, improving asset processing efficiency for designers and large-scale build workflows

3D Gameplay Audio Programmer

REMOTE

Serious Point Games Ltd

March 2024 - May 2025

- **Integrated** Wwise with Unity to implement gameplay audio systems, including footsteps, interaction audio, ambience transitions, and state-driven runtime event pipelines
- **Used** Wwise Rooms & Portals to build spatial audio behavior with room-aware reverb, environmental transitions, and dynamic propagation across connected spaces

Full Stack Engineer

BALTIMORE MD

Quest2Learn LLC

December 2023 - October 2024

- **Developed** full-stack interactive systems using React Three Fiber (R3F), supporting a scalable AR learning platform across mobile and web.
- **Designed and implemented** backend API integrations and migration pipelines, transitioning 40% of legacy Unity content into web-based systems

PROJECTS

Deja You – Technical Lead | Built gameplay-driven DSP for time-loop ghosts using real-time filtering, delay, bit-depth reduction, gain shaping, and a native C++ per-sample DSP kernel

Ducks Afar – Project Lead | Developed gameplay architecture and UI/audio feedback systems for a \$5K grant-funded narrative puzzle game built around spatial arithmetic and resource transformation

TECHNICAL SKILLS

Programming Languages | C/C++, C#, Python, Java, TypeScript/JavaScript

Low-Level & Systems | Multithreading, Memory Optimization, DSP, Runtime Performance, Data-Oriented Design, Profiling

Graphics & Engines | Unity, Unreal Engine 5, GLSL, HLSL

Tools & Middleware | Wwise, FMOD, .NET/WPF, Reaper, Logic Pro, Pro Tools